TRIGONOMETRY

त्रिकोणमिति

SHEET-04

CLASS NOTES

CONCEPT OF VALUE PUTTING

1. The value of $(1+\sin^4 A - \cos^4 A)\operatorname{cosec}^2 A$ is:

SSC CPO 03/10/2023 (Shift-02)

(a) -1 (b) 1

(c) -2 (d) 2

$\theta = 45°$ $\left[1 + \left(\frac{1}{\sqrt{2}}\right)^4 - \left(\frac{1}{\sqrt{2}}\right)^4\right] (\sqrt{2})^2$

$= 1 \times 2$

$= 2$

2. $(\cot\theta + \tan\theta)(\text{cosec}\,\theta - \sin\theta)(\cos\theta - \sec\theta)$ = ----.

SSC CHSL 03/07/2024 (Shift-01)

(a) 0 (b) 1

(c) 2 (d) –1

$\theta = 45^\circ$

$$(1+1)\left(\sqrt{2}-\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}-\sqrt{2}\right)$$

$$= 2 \times \frac{1}{\sqrt{2}} \times \frac{(-1)}{\sqrt{2}}$$

$$= -1$$

3. **Find the value of**

$(\text{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta)$.

SSC CGL 25/09/2024 (Shift-01)

(a) 0 (b) $\frac{1}{2}$

(c) 1 (d) −1

$\theta = 45°$

$\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)\left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)(1+1)$

$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times 2$

$= 1$

4. The value of

$$\left(\frac{\sin A}{1-\cos A}+\frac{1-\cos A}{\sin A}\right)\div\left(\frac{\cot^2 A}{1+\operatorname{cosec} A}+1\right)$$ is:

SSC CGL TIER-II 12/09/2019

(a) $\frac{3}{2}$ (b) $\frac{1}{2}$

(c) 1 (d) 2

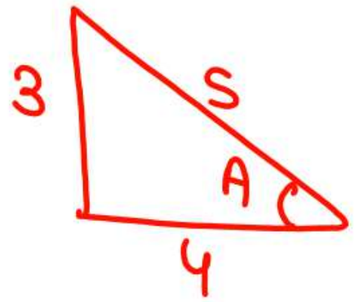
$$\left(\frac{\frac{3}{5}}{\frac{1-4}{5}}+\frac{\frac{1-4}{5}}{\frac{3}{5}}\right)\div\left(\frac{\frac{16}{9}}{\frac{1+5}{3}}+1\right)$$

$$=\left(\frac{\frac{3}{5}}{\frac{1}{5}}+\frac{\frac{1}{5}}{\frac{3}{5}}\right)\div\left(\frac{\frac{16}{9}}{\frac{8}{3}}+1\right)$$

$$=\left(3+\frac{1}{3}\right)\div\left(\frac{2}{3}+1\right)$$

$$=\frac{10}{3}\div\frac{5}{3}$$

$$=2$$



$A = 90^\circ$

$C = 0 \quad S = 1$

4. The value of

$$\left(\frac{\sin A}{1-\cos A}+\frac{1-\cos A}{\sin A}\right)\div\left(\frac{\cot^2 A}{1+\operatorname{cosec} A}+1\right)$$ **is:**

SSC CGL TIER-II 12/09/2019

(a) $\frac{3}{2}$ **(b)** $\frac{1}{2}$

(c) 1 **(d)** 2

$$\left(\frac{1}{1}+\frac{1}{1}\right)\div\left(\frac{0}{1+1}+1\right)$$

$$= 2 \div 1$$

$$= 2$$

5. The value of

$$\frac{\sin A}{\cot A + \operatorname{cosec} A} - \frac{\sin A}{\cot A - \operatorname{cosec} A} - 1$$

SSC CPO 11/11/2022 (Shift-02)

(a) $\frac{1}{2}$ (b) 1

(c) 0 (d) 2

$A = 90°$

$C = 0 \quad S = 1$

$$\frac{1}{0+1} - \frac{1}{0-1} - 1$$

$$= 1 + 1 - 1$$

$\left(\frac{1}{\sqrt{2}}\right)^6 = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$

$= \frac{1}{8}$

6. The expression $(\cos^6\theta + \sin^6\theta - 1)(\tan^2\theta + \cot^2\theta + 2) + 3$ is equal to:

SSC CGL 13/04/2022 (Shift-02)

(a) 0 (b) – 1

(c) 2 (d) 1

$\theta = 45^\circ$

$\left[\left(\frac{1}{\sqrt{2}}\right)^6 + \left(\frac{1}{\sqrt{2}}\right)^6 - 1\right](1+1+2) + 3$

$= \left(\frac{1}{8} + \frac{1}{8} - 1\right) \times 4 + 3$

$= \left(\frac{1+1-8}{8}\right) \times 4 + 3$

$= \frac{-6}{8} \times 4 + 3$

$= 0$

7. Let $0° < \theta < 90°$, $(1 + \cot^2\theta)(1 + \tan^2\theta) \times (\sin\theta - \operatorname{cosec}\theta)(\cos\theta - \sec\theta)$ is equal to:

SSC CGL TIER-II 29/01/2022

(a) $\sec\theta\operatorname{cosec}\theta$ (b) $\sec\theta + \operatorname{cosec}\theta$

(c) $\sin\theta + \cos\theta$ (d) $\sin\theta\cos\theta$

$\theta = 45°$

$2 \times 2 \times \left(\frac{1}{\sqrt{2}} - \sqrt{2}\right)\left(\frac{1}{\sqrt{2}} - \sqrt{2}\right)$

$= 4\left(-\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{2}}\right)$

$= 4 \times \frac{1}{2} = 2$

$\theta = 45$

8. **The expression**

$$\frac{\tan^6\theta - \sec^6\theta + 3\sec^2\theta\tan^2\theta}{\tan^2\theta + \cot^2\theta + 2}$$ **$0° < \theta < 90°$, is equal to:**

SSC CGL TIER-II 03/02/2022

~~(a)~~ $\sec^2\theta\operatorname{cosec}^2\theta$ ~~(b)~~ $-\sec^2\theta\operatorname{cosec}^2\theta$

~~(c)~~ $\cos^2\theta\sin^2\theta$ ~~(d)~~ $-\cos^2\theta\sin^2\theta$

$$\frac{1-(\sqrt{2})^6+3\times2\times1}{4}$$

$$= \frac{1-8+6}{4} = -\frac{1}{4}$$

$$-\frac{1}{2}\times\frac{1}{2} = -\frac{1}{4}$$

9. The expression

$$\frac{(1-\sin\theta+\cos\theta)^2(1-\cos\theta)\sec^3\theta\,\mathrm{cosec}^2\theta}{(\sec\theta-\tan\theta)(\tan\theta+\cot\theta)}, \quad 0^\circ < \theta < 90^\circ,$$

is equal to

SSC CGL TIER-II 03/02/2022

(a) $\sin\theta$ **(b)** $2\cos\theta$

(c) $\cot\theta$ **(d)** $2\tan\theta = 2$

$$1 \times \frac{\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right) \times 2\sqrt{2} \times 2}{(\sqrt{2}-1) \times 2} = 2$$

$\theta = 45^\circ$

$\sin\theta = \cos\theta$

$\sec\theta = \mathrm{cosec}\,\theta$

$\tan\theta = \cot\theta$

10. The expression

$$\left\{\frac{(1-2\sin^2\theta\cos^2\theta)(\cot\theta+1)\cos\theta}{(\sin^4\theta+\cos^4\theta)(1+\tan\theta)\operatorname{cosec}\theta}-1\right\},\ 0^\circ < \theta < 90^\circ,$$ **is equal to:**

SSC CGL TIER-II 03/02/2022

(a) $\cos^2\theta$
(b) $-\sin^2\theta$
(c) $\sec^2\theta$
(d) $-\sec^2\theta$

$\theta = 45^\circ$

$$\frac{\left(1-2\times\frac{1}{2}\times\frac{1}{2}\right)\times 2\times\frac{1}{\sqrt{2}}}{\left(\frac{1}{4}+\frac{1}{4}\right)\times 2\times\sqrt{2}}-1$$

$$= \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}}}\times\frac{1}{2}-1 = -\frac{1}{2}$$

P = Q = 1

∴ cot α = 1

α = 45°

11. If $\cot\alpha = \frac{P}{Q}$, then find the value of

$$\frac{P\cos\alpha - Q\sin\alpha}{P\cos\alpha + Q\sin\alpha} - \frac{P^2 - Q^2}{P^2 + Q^2} + 3.$$

• **SSC CHSL 09/07/2024 (Shift-01)**

(a) $\frac{P^2}{P^2+Q^2} = \frac{1}{2}$ (b) $\frac{P}{Q}$ 1

(c) 0 (d) 3

$\frac{C-S}{C+S} - \frac{0}{0} + 3 =$

12. If $m = a\sec A$ and $y = b\tan A$, then find the value of

$$b^2m^2 - a^2y^2 + \frac{a^2y^2}{b^2m^2} + \cos^2 A.$$

SSC CGL 13/09/2024 (Shift-03)

(a) a^2b^2 1

(b) $1-a^2b^2$ 0

(c) $a^2b^2 + 2$ 3

(d) $a^2b^2 + 1$ 2

$\sqrt{2} = \sec A$ | $1 = \tan A$

$A = 45°$ | $A = 45°$

$1 \times 2 - 1 \times 1 + \frac{1}{2} + \frac{1}{2}$

$= 1 + 1$

$= 2$

13. **If $m = \sec\theta - \tan\theta$ and $n = \operatorname{cosec}\theta + \cot\theta$, then what is the value of $m + n(m - 1)$?**

SSC CHSL 13/03/2023 (Shift-02)

(a) 2 **(b) 1**

(c) 0 **(d) −1**

$\theta = 45°$, $m = \sqrt{2} - 1$ $n = \sqrt{2} + 1$

$= m + nm - n$

$= \sqrt{2} - 1 + 2 - 1 - \sqrt{2} - 1$

$= -1$

14. **Which of the following is true when $x = \sin A + \cos A$; $y = \sec A + \operatorname{cosec} A$?**

SSC CPO 27/06/2024 (Shift-03)

(a) $y(1 + x^2) = 2x$ (b) $y - 2x = x^2y$

(c) $y + 2x = x^2y$ (d) $y(1 - 2x^2) = x$

$A = 45°$, $x = \frac{1}{\sqrt{2}} \times \sqrt{2}$ $y = 2\sqrt{2}$

(b) $2\sqrt{2}(1+2)$

$2\sqrt{2} \times 3$

$6\sqrt{2} \neq 2\sqrt{2}$

(d) $2\sqrt{2}(1-2\times2)$

(c) $2\sqrt{2}+2\sqrt{2}$ $2 \times 2\sqrt{2}$

$4\sqrt{2} = 4\sqrt{2}$

$A = 30°$

$\sqrt{3} + \frac{\sqrt{3}}{2} = P$ | $\sqrt{3} - \frac{\sqrt{3}}{2} = Q$

$\Rightarrow \frac{3\sqrt{3}}{2} = P$ | $\frac{\sqrt{3}}{2} = Q$

© $\frac{2}{4} \times \frac{3}{2} = \frac{27}{4} - \frac{3}{4}$

$6 = \frac{24}{4} 6$

15. If cotA + cosA = p and cot A – cosA = q. then which of the following relation is correct.?

SSC CHSL 03/08/2023 (Shift-04)

(a) $\sqrt{\frac{1}{16pq}} = p^2 + q^2$

(b) $\frac{1}{4pq} = p^2 + q^2$

(c) $\sqrt{16pq} = p^2 - q^2$

(d) $4pq = p^2 + q^2$

$PQ = \frac{3\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}$

$= \frac{3 \times 3}{4} = \frac{9}{4}$

15. If cotA + cosA = p and cot A – cosA = q. then which of the following relation is correct.?

SSC CHSL 03/08/2023 (Shift-04)

(a) $\sqrt{\frac{1}{16pq}} = p^2 + q^2$ **(b)** $\frac{1}{4pq} = p^2 + q^2$

(c) $\sqrt{16pq} = p^2 - q^2$ **(d)** $4pq = p^2 + q^2$

$(\cot\theta + \cos A)^2 = p^2$

$-\quad (\cot\theta - \cos A)^2 = q^2$

$\Rightarrow 4\cot\theta \cdot \cos A = p^2 - q^2$

$\Rightarrow 4\sqrt{pq} = p^2 - q^2$

$(a+b)^2 - (a-b)^2 = 4ab$

$(\cot A + \cos A) \times (\cot A - \cos A) = p \times q$

$\cot^2 A - \cos^2 A = pq$

$\Rightarrow \frac{c^2}{s^2} - c^2 = pq$

$\Rightarrow \frac{c^2}{s^2} \times c^2 = pq$

$\Rightarrow \cot A \cos A = \sqrt{pq}$

$A = 60^\circ$

$\sqrt{3} + \frac{\sqrt{3}}{2} = x$ | $\sqrt{3} - \frac{\sqrt{3}}{2} = y$

$\frac{3\sqrt{3}}{2} = x$ | $\frac{\sqrt{3}}{2} = y$

$\therefore x^2 - y^2 = \frac{27}{4} - \frac{3}{4} = 6$

16. If tanA + sinA = x, tanA – sinA = y, what is the value of $x^2 - y^2$?

SSC CHSL 11/07/2024 (Shift-04)

(a) $2\sqrt{\frac{x}{y}}$ $= 2\sqrt{\frac{\frac{3\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}}}$

(b) $4\sqrt{\frac{x}{y}}$

(c) $\sqrt{xy}$

(d) $4\sqrt{xy}$ $= 4\sqrt{\frac{3\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}}$

$= 4 \times \frac{3}{2}$

17. If $(\sin x + \sin y) = a$ and $(\cos x + \cos y) = b$, then find the value of $(\sin x \sin y + \cos x \cos y)$.

SSC CPO 27/06/2024 (Shift-02)

(a) $\frac{a^2 + b^2 - 1}{2} = \frac{3}{2}$

(b) $\frac{a^2 + b^2 - 1}{2} = \frac{3}{2}$

(c) $\frac{a^2 + b^2 - 2}{2} = \frac{2}{2}$

(d) $\frac{a^2 + b^2 + 2}{2} = \frac{6}{2}$

$x = 0°$ $y = 0°$

$0 + 0 = a$ $\quad 1 + 1 = b$

$a = 0$, $b = 2$

$0 + 1 = 1$

18. **If $a\sin^3 X + b\cos^3 X = \sin X \cos X$ and $a\sin X = b\cos X$, then find the value of $a^2 + b^2$, provided that X is neither 0° nor 90°.**

SSC CGL 11/09/2024 (Shift-01)

(a) 0 **(b) 1**

(c) $a^2 - b^2$ **(d) $a^2 + b^2$**

Basic

$a\sin x = b\cos x$ —— (i)

$a\sin^3 x + b\cos^3 x = \sin x \cos x$

$\Rightarrow a\sin x \cdot \sin^2 x + b\cos^3 x = \sin x \cos x$

$\Rightarrow b\cos x \cdot \sin^2 x + b\cos^3 x = \sin x \cos x$ ($\cos^2 x$)

$\Rightarrow b(\sin^2 x + \cos^2 x) = \sin x$

$\Rightarrow b = \sin x$

put (b) in eqn (i)

$a\sin x = \sin x \cos x$

$a = \cos x$

$a^2 + b^2$

$= \cos^2 x + \sin^2 x$

$= 1$

18. If $a\sin^3 X + b\cos^3 X = \sin X \cos X$ and $a\sin X = b\cos X$, then find the value of $a^2 + b^2$, provided that X is neither 0° nor 90°.

SSC CGL 11/09/2024 (Shift-01)

(a) 0
(b) 1
(c) $a^2 - b^2$
(d) $a^2 + b^2$

$X = 45°$

$a \times \frac{1}{\sqrt{2}} = b \times \frac{1}{\sqrt{2}}$

$a = b$

$2 \times a \sin^3 x = \sin^2 x$

$\Rightarrow 2 \times a \times \frac{1}{\sqrt{2}} = 1$

$a = \frac{1}{\sqrt{2}}$

$a^2 + b^2 = \frac{1}{2} + \frac{1}{2} = 1$

0, 30, 45, 60, 90

19. $[(\sin\theta \tan\theta + \cos\theta)^2 - 1]$ is equal to:

SSC Phase-X 02/08/2022 (Shift-02)

(a) $\sec^2\theta$ (b) $\sec\theta$

(c) $\operatorname{cosec}\theta$ (d) $\tan^2\theta$ 1

$\theta = 45^\circ$

$$\left(\frac{1}{\sqrt{2}} \times 1 + \frac{1}{\sqrt{2}}\right)^2 - 1$$

$$= \left(\frac{\sqrt{2}}{\sqrt{2}}\right)^2 - 1$$

$$= 2 - 1$$

$$= 1$$

$\cos 2\theta \rightarrow 1-2\sin^2\theta$, $2\cos^2\theta-1$

20. The value of

$$\left(\frac{\sin\theta-2\sin^3\theta}{2\cos^3\theta-\cos\theta}\right)^3 \frac{1}{\tan\theta}-\sec^2\theta \text{ is:}$$

SSC CGL 12/09/2024 (Shift-01)

(a) 2 **(b) 1**

(c) 0 **(d) −1**

$$\left[\frac{S(1-2S^2)}{C(2C^2-1)}\right]^3 \times \cot\theta-\sec^2\theta$$

$$= \tan^3\theta \times \cot\theta-\sec^2\theta$$

$$= 1\times 1-2$$

$$= -1$$

$\theta = 45^\circ$

21. $\dfrac{\cot^3\theta}{\operatorname{cosec}^2\theta} + \dfrac{\tan^3\theta}{\sec^2\theta} + 2\sin\theta\cos\theta = ?$

SSC CGL 16/08/2021 (Shift-02)

(a) $\operatorname{cosec}\theta\sec\theta$ 2

(b) $\operatorname{cosec}^2\theta\sec^2\theta$ 4

(c) $\sin\theta\cos\theta$ $\frac{1}{2}$

(d) $\sin\theta$ $\frac{1}{\sqrt{2}}$

$\frac{1}{2} + \frac{1}{2} + 2 \times \frac{1}{2}$

$= 1 + 1$

$= 2$

$\sec^2\theta + \operatorname{cosec}^2\theta = \sec^2\theta \times \operatorname{cosec}^2\theta$

$\tan\theta + \cot\theta = \sec\theta \times \operatorname{cosec}\theta$

22. $\frac{1+\cos\theta-\sin^2\theta}{\sin\theta(1+\cos\theta)} \times \frac{\sqrt{\sec^2\theta+\operatorname{cosec}^2\theta}}{\tan\theta+\cot\theta}$, $0° < \theta < 90°$

SSC CGL TIER-II 29/01/2022

(a) $\tan\theta$ (b) $\sec\theta$

(c) $\operatorname{cosec}\theta$ (d) $\cot\theta$

$$\frac{C^2+C}{S(1+C)} \times \frac{\sec\theta\times\operatorname{cosec}\theta}{\sec\theta\times\operatorname{cosec}\theta}$$

$$= \frac{C(C+1)}{S(1+C)}$$

$$= \cot\theta$$

$\theta = 45^\circ$

23. The expression

$$\frac{\cos^4\theta - \sin^4\theta + 2\sin^2\theta + 3}{(\operatorname{cosec}\theta + \cot\theta + 1)(\operatorname{cosec}\theta - \cot\theta + 1) - 2}$$

$0^\circ < \theta < 90^\circ$, **is equal to**

SSC CGL TIER-II 03/02/2022

(a) $\frac{1}{2}\sin\theta$ **(b)** $2\sin\theta$

(c) $\sec\theta$ **(d)** $2\operatorname{cosec}\theta$

$\sqrt{2} \times \frac{1}{\sqrt{2}}$

$$\frac{2 \times \frac{1}{2} + 3}{(\sqrt{2}+2)\times\sqrt{2} - 2} = \frac{4}{2 + 2\sqrt{2} - 2} = \frac{4}{2\sqrt{2}} = \sqrt{2}$$

$\cos^4\theta - \sin^4\theta$

$= (\cos^2\theta - \sin^2\theta)(\cos^2\theta + \sin^2\theta)$

$= (\cos^2\theta - \sin^2\theta) \times 1$

$= 1 - \sin^2\theta - \sin^2\theta$

$= 1 - 2\sin^2\theta$

* $\cos^4\theta - \sin^4\theta = 1 - 2\sin^2\theta$

* $a^2 - b^2 = (a-b)(a+b)$
* $a^4 - b^4 = (a^2 - b^2)(a^2 + b^2)$
* $a^6 - b^6 = (a^3 - b^3)(a^3 + b^3)$
* $a^8 - b^8 = (a^4 - b^4)(a^4 + b^4)$

$$\frac{1-25^2+25^2+3}{\left(\frac{5}{3}+\frac{4}{3}+1\right)\left(\frac{5}{3}-\frac{4}{3}+1\right)-2}$$

$$= \frac{4}{4\times\frac{4}{3}-2} = \frac{4}{\frac{10}{3}} = \frac{12}{10} = \frac{6}{5}$$

23. The expression

$$\frac{\cos^4\theta-\sin^4\theta+2\sin^2\theta+3}{(\operatorname{cosec}\theta+\cot\theta+1)(\operatorname{cosec}\theta-\cot\theta+1)-2}$$

$0^\circ < \theta < 90^\circ$, **is equal to**

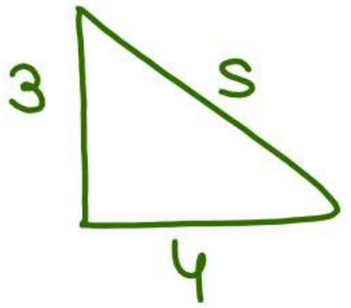
SSC CGL TIER-II 03/02/2022

(a) $\frac{1}{2}\sin\theta = \frac{1}{2}\times\frac{3}{5}$

(b) $2\sin\theta$ $\quad 2\times\frac{3}{5}$

(c) $\sec\theta$

(d) $2\operatorname{cosec}\theta$ $\quad 2\times\frac{5}{3}$



$$\frac{4}{[(\operatorname{cosec}\theta+1)+\cot\theta][(\operatorname{cosec}\theta+1)-\cot\theta]-2}$$

$$= \frac{4}{(\operatorname{cosec}\theta+1)^2-\cot^2\theta-2}$$

$$= \frac{4}{\operatorname{cosec}^2\theta+1+2\operatorname{cosec}\theta-\cot^2\theta-2}$$

$$= \frac{2\cdot 4}{1+1+2\operatorname{cosec}\theta-2}$$

$$= 2\sin\theta$$

23. The expression

$$\frac{\cos^4\theta-\sin^4\theta+2\sin^2\theta+3}{(\operatorname{cosec}\theta+\cot\theta+1)(\operatorname{cosec}\theta-\cot\theta+1)-2}$$

$0^\circ < \theta < 90^\circ$, is equal to

SSC CGL TIER-II 03/02/2022

(a) $\frac{1}{2}\sin\theta$ **(b)** $2\sin\theta$

(c) $\sec\theta$ **(d)** $2\operatorname{cosec}\theta$

24. **Using trigonometric formulas, find the value of**

$$\left(\frac{\sin(x-y)}{\sin(x+y)}\right)\left(\frac{\tan x+\tan y}{\tan x-\tan y}\right)$$

SSC CGL 18/07/2023 (Shift-01)

(a) –2 **(b) 2**

(c) 0 **(d) 1**

$y = 0$ $\tan y = 0$

1×1

$= 1$

$A = 30$ | $B = 60^\circ$

$\frac{1}{\sqrt{3}} = a \times \sqrt{3}$ | $\frac{1}{2} = b \times \frac{\sqrt{3}}{2}$

$\frac{1}{3} = a$ | $\frac{1}{\sqrt{3}} = b$

$\therefore \cos^2 30^\circ = \frac{3}{4}$

25. If $\tan A = a \tan B$ and $\sin A = b \sin B$, then the value of $\cos^2 A$ is:

SSC CHSL 05/07/2024 (Shift-01)

$\frac{3}{4}$ (a) $\frac{b^2-1}{a^2-1}$ $\frac{\frac{1}{3}-1}{\frac{1}{9}-1}$

(b) $\frac{a^2-1}{b^2-1}$

(c) $\frac{b^2+1}{a^2+1}$

(d) $\frac{a^2+1}{b^2+1}$

$\frac{3-9}{1-9} = \frac{-6}{-8} \frac{3}{4}$

$A = 60^\circ$ $B = 30^\circ$

$\sqrt{3} = a \times \frac{1}{\sqrt{3}}$ | $\frac{\sqrt{3}}{2} = b \times \frac{1}{2}$

$3 = a$ | $\sqrt{3} = b$

$\cos^2 60^\circ = \frac{1}{4}$

25. If tanA = a tanB and sinA = b sinB, then the value of $\cos^2 A$ is:

SSC CHSL 05/07/2024 (Shift-01)

(a) $\frac{b^2-1}{a^2-1}$ $\frac{3-1}{9-1} = \frac{2}{8}$ $\frac{1}{4}$ **(b)** $\frac{a^2-1}{b^2-1}$

(c) $\frac{b^2+1}{a^2+1}$ **(d)** $\frac{a^2+1}{b^2+1}$

Basic

$$\frac{1-2SC}{1+2SC} \times \frac{C+S}{C-S}$$

$$= \left(\frac{C^2+S^2-2SC}{C^2+S^2+2SC}\right) \times \frac{(C+S)}{(C-S)}$$

$$= \frac{(C-S)^2}{(C+S)^2} \times \frac{(C+S)}{(C-S)}$$

$$= \frac{C-S}{C+S} = \frac{1-\tan t}{1+\tan t}$$

26. The value of the expression

$$\frac{1-\sin(2t)}{1+\sin(2t)} \times \frac{\cos(t)+\sin(t)}{\cos(t)-\sin(t)}$$ **is:**

SSC CGL 19/09/2024 (Shift-02)

(a) $\frac{1-2\tan(t)}{1+2\tan(t)}$

(b) $\frac{1-\tan(t)}{1+\tan(t)}$

(c) $\frac{1+2\tan(t)}{1-2\tan(t)}$

(d) $\frac{1+\tan(t)}{1-\tan(t)}$

$\sin 2\theta = 2\sin\theta\cos\theta$

$S^2+C^2=1$

or

$C^2+S^2=1$

$\text{Cosec } 75^\circ = \frac{H}{P} = \frac{2\sqrt{2}}{(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)}$

$= \frac{2(\sqrt{6}-\sqrt{2})}{2}$

27. Using $\text{cosec}(\alpha+\beta)$

$$\text{Cosec}(\alpha+\beta) = \frac{\sec\alpha \times \sec\beta \times \text{cosec}\alpha \times \text{cosec}\beta}{\sec\alpha \times \text{cosec}\beta + \text{cosec}\alpha \times \sec\beta},$$

find the value of cosec 75°.

SSC CGL 25/07/2023 (Shift-01)

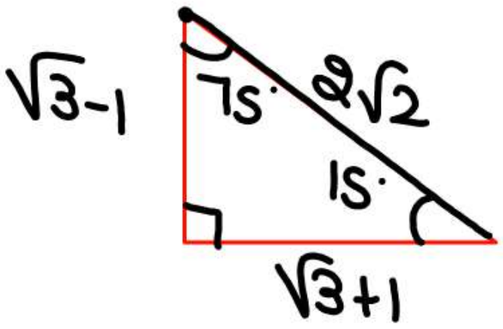
(a) $1\frac{\sqrt{6}+\sqrt{2}}{4}$

(b) $\frac{\sqrt{6}-\sqrt{2}}{4}$

(c) $\sqrt{6}-\sqrt{2}$

(d) $\sqrt{6}+\sqrt{2}$

$3-1$



$\theta = 0^\circ$

28. The value of the expression

$4(\sin^6 A + \cos^6 A) - 6(\sin^4 A + \cos^4 A) + 8$ is:

SSC CHSL 02/08/2023 (Shift-04)

(a) 4 **(b) 8**

(c) 7 **(d) 6**

$4(0+1) - 6(0+1) + 8$

$= 4 - 6 + 8$

$= 6$

29. Find the value of the following expression.

$12(\sin^4\theta + \cos^4\theta) + 18(\sin^6\theta + \cos^6\theta) + 78\sin^2\theta\cos^2\theta$

SSC CGL 18/07/2023 (Shift-03)

(a) 30 (b) 40

(c) 10 (d) 20

$12(0+1) + 18(0+1) + 78 \times 0 \times 1$

$= 12 + 18 + 0$

$= 30$

$\theta = 0^\circ$

$$\frac{2\times 0\,(1+0)}{1+0+1} = \frac{0}{2} = 0$$

30. $\dfrac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$ is equal to :

SSC CGL TIER-II 11/09/2019

0 (a) $1 - \cos A$ 1

(b) 1

2 (c) $1 + \cos A$ 1

(d) 1

31. **Find the value of the following expression $5(\sin^4\theta + \cos^4\theta) + 3(\sin^6\theta + \cos^6\theta) + 19\sin^2\theta \cos^2\theta$.**

SSC CGL 07/12/2022 (Shift-03)

(a) 8
(b) 5
(c) 6
(d) 7

$5(1+0)+3(1+0)+19\times0\times1$

$= 5+3+0$

$= 8$

$\theta = 0^\circ$

$\sin 0^\circ = 0$

$\cos 0^\circ = 1$

32. The value of

$$\frac{\sec^6\theta - \tan^6\theta - 3\sec^2\theta\tan^2\theta + 1}{\cos^4\theta - \sin^4\theta + 2\sin^2\theta + 2}$$ **is:**

SSC CGL 05/03/2020 (Shift-01)

(a) $\frac{2}{3}$

(b) 1

(c) $\frac{3}{4}$

(d) $\frac{1}{2}$

$$\frac{1 - 0 - 3 \times 1 \times 0 + 1}{1 - 0 + 0 + 2}$$

$$= \frac{1+1}{1+2} = \frac{2}{3}$$

33. What is the value of $\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$?

SSC CPO 29/06/2024 (Shift-03)

(a) 0
(b) 1
(c) −1
(d) 2

$x = 90°$ $y = 0°$

$$\frac{(1+0)(0-1)}{(0+1)(1-0)}$$

$$= \frac{1(-1)}{1 \times 1} = \frac{-1}{1}$$

$\operatorname{cosec}^2\theta - 1 = 1 - e^2$

$\Rightarrow \operatorname{cosec}^2\theta = 2 - e^2$

34. If $\cot^2\theta = 1 - e^2$, then the value of $\operatorname{cosec}\theta + \cot^3\theta \sec\theta$ is:

SSC CPO 05/10/2023 (Shift-03)

(a) $(2 - e^2)^{\frac{3}{2}}$

(b) $(1 - e^2)$

(c) $(1 - e^2)^{\frac{3}{2}}$

(d) $(2 - e^2)^{\frac{1}{2}}$

$$\frac{1}{S} + \frac{c^2}{S^3} \times \frac{1}{c}$$

$$= \frac{S^2 + c^2}{S^3} = \frac{1}{S^3} = \operatorname{cosec}^3\theta$$

$\operatorname{cosec}\theta = (2 - e^2)^{\frac{1}{2}}$

$\therefore \operatorname{cosec}^3\theta = (2 - e^2)^{\frac{3}{2}}$

$\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$\Rightarrow \operatorname{cosec}^2\theta - 1 = \cot^2\theta$

$\theta = 45^\circ$

$1 = 1 - e^2$

$\Rightarrow e = 0$

34. If $\cot^2\theta = 1 - e^2$, then the value of $\text{cosec}\,\theta + \cot^3\theta\,\sec\theta$ is:

SSC CPO 05/10/2023 (Shift-03)

(a) $(2)^{\frac{3}{2}}$ (b) (1)

(c) $(1)^{\frac{3}{2}}$ (d) $(2)^{\frac{1}{2}}$

$\sqrt{2} + 1 \times \sqrt{2}$

$= 2\sqrt{2}$

$= 2^1 \times 2^{\frac{1}{2}}$

$= 2^{1+\frac{1}{2}} = 2^{\frac{3}{2}}$

$\theta = 45^\circ$

$1 = 1 - a^2$

$\Rightarrow a^2 = 0$

$\therefore a = 0$

35. If $\tan^2\theta = 1 - a^2$, then the value of $\sec\theta + \tan^3\theta$ $\text{cosec}\theta$ is:

SSC CGL 01/12/2022 (Shift-04)

(a) $(2-a)^{\frac{3}{2}}$

(b) $(a^2-1)^{\frac{3}{2}}$

(c) $(2-a^2)^{\frac{3}{2}}$

(d) $a^{\frac{3}{2}}$

$\sec\theta + \tan^3\theta \cdot \text{cosec}\theta$

$= \sqrt{2} + \sqrt{2}$

$= 2\sqrt{2}$

$= 2^{\frac{3}{2}}$

35. If $\tan^2\theta = 1 - a^2$, then the value of $\sec\theta + \tan^3\theta$ $\operatorname{cosec}\theta$ is:

SSC CGL 01/12/2022 (Shift-04)

(a) $(2-a)^{\frac{3}{2}}$

(b) $(a^2-1)^{\frac{3}{2}}$

(c) $(2-a^2)^{\frac{3}{2}}$

(d) $a^{\frac{3}{2}}$

$\sec^2\theta - 1 = 1 - a^2$

$\Rightarrow \sec^2\theta = (2-a^2)^1$

$\therefore \sec\theta = (2-a^2)^{\frac{1}{2}}$

$\frac{1}{C} + \frac{S^2}{C^3} \times \frac{1}{S}$

$= \frac{C^2+S^2}{C^3} = \frac{1}{C^3} = \sec^3\theta$

$\therefore \sec^1\theta = (2-a^2)^{\frac{1}{2}}$

$\sec^3\theta = (2-a^2)^{\frac{3}{2}}$

$\alpha = 45^\circ$

36. If $\tan^2\alpha = 3 + Q^2$, then $\sec\alpha + \tan^3\alpha \operatorname{cosec}\alpha = ?$

SSC CGL 07/12/2022 (Shift-01)

(a) $(3 + Q^2)^{\frac{3}{2}}$ $= 1^{\frac{3}{2}}$

(b) $(7 + Q^2)^{\frac{3}{2}}$ $5^{\frac{3}{2}}$

(c) $(5 - Q^2)^{\frac{3}{2}}$ $7^{\frac{3}{2}}$

(d) $(4 + Q^2)^{\frac{3}{2}}$ $2^{\frac{3}{2}}$

$1 = 3 + Q^2$

$\therefore -2 = Q^2$

$\sqrt{2} + \sqrt{2}$

$= 2\sqrt{2}$

$= 2^{\frac{3}{2}}$